

Title: Vulnerability Assessment of Networked Systems

Context: Academic Lab Environment

Executive Summary:

This project involved conducting a structured vulnerability assessment against hidden systems within a controlled lab environment. Using industry-standard tools including Nmap, ARP-Scan, and the Greenbone Vulnerability Manager, I performed host discovery, service enumeration, and vulnerability analysis to evaluate the security posture of identified systems.

One hidden Linux system was successfully discovered and assessed, revealing multiple high- and medium-severity vulnerabilities related to outdated services, weak cryptographic configurations, insecure authentication mechanisms, and exposed network services. A second hidden Windows system could not be located despite exhaustive discovery attempts, which was documented and reported in alignment with provided guidance.

Findings were categorized by severity using CVSS-based scoring and accompanied by remediation recommendations focused on patching, service hardening, cryptographic modernization, and access restriction. This assessment emphasized risk prioritization, evidence documentation, and practical mitigation strategies consistent with vulnerability management best practices.